Geol Rundsch (1996) 85:293–302 © Springer-Verlag 1996

ORIGINAL PAPER

H. C. Soffel · M. Davoudzadeh · C. Rolf · S. Schmidt

New palaeomagnetic data from Central Iran and a Triassic palaeoreconstruction

Received: 28 February 1995 / Accepted: 22 November 1995

Abstract New pole positions for Triassic and Cretaceous times have been obtained from volcanic and sedimentary sequences in Central Iran. These new results confirm the general trend of the Apparent Polar Wander Path (APWP) of the Central-East-Iran microplate (CEIM) from the Triassic through the Tertiary as published by Soffel and Förster (1983, 1984). Two new palaeopoles for the Triassic of the CEIM have been obtained; limestones and tuffs from the Nakhlak region yield a mean direction of 094.0°/25.0°, N = 12, k = 4.1, $\alpha_{95} = 24.7°$, after bedding correction, corresponding to a palaeopole position of 310.8°E; 3.9°S, and volcanic rocks from the Sirjan regions yield a mean direction of 114.5°/35.1°, N = 44, k = 45.9, $\alpha_{95} = 3.2°$ after bedding correction and a palaeopole position of 295.8°E; 10.3°N. Combining these with the two previously published results yields a new palaeopole position of 317.5°E; 12.7°N, for the Triassic of the CEIM, thus confirming that large counterclockwise rotations of the CEIM have occurred since the Triassic time. New results have also been obtained from Cretaceous limestones from the Saghand region of the CEIM. The mean direction of 340.7°/26.3°, N = 33, k = 44.3, $\alpha_{95} = 3.8°$, and the corresponding palaeopole position of 283.1°E; 64.4°N, is in agreement with previously determined Cretaceous palaeopole positions of the CEIM. Furthermore, results have also been obtained from Triassic dolomite, limestone, sandstone and siltstone from the Natanz region, which is located to the west of the CEIM. A total of 161 specimens from 44 cores taken at five sites gave a mean direction of the five sites at 033.3°/25.1°, N = 5, k = 69.0, $\alpha_{95} = 9.3°$ and a palaeopole position of 167.2°E; 53.7°N. They pass the positive fold test of McElhinny (1964) on the level of 99% confidence. This pole position is in fairly good agreement with the mean Triassic pole position of the Turan Plate (149°E; 49°N). It indicates that the area of Natanz has not undergone the large counterclockwise rotation relative to the Turan plate since the Triassic, which has been shown for the CEIM. A Triassic palaeogeographic reconstruction of Iran, Arabia (Gondwana) and the Turan Plate (Eurasia) is also presented.

Key words Iran · palaeomagnetism · Triassic · Cretaceous · Central Iran · geotectonics · palaeoreconstruction

H. C. Soffel (✉) · S. Schmidt
Institut für Allgemeine und Angewandte Geophysik, Universität München, Theresienstrasse 41, D-80333 München, Germany

M. Davoudzadeh
Institut für Allgemeine und Angewandte Geologie, Universität München, Luisenstrasse 37, D-80333 München, Germany

C. Rolf
Niedersächsisches Landesamt für Bodenforschung, Aussenstelle Grubenhagen, OT Rotenkirchen, D-37574 Einbeck, Germany

Introduction

The geotectonic position of Iran with respect to the Scytho-Turanian plate (Eurasia) in the north and the Arabian plate (Gondwana) in the south is controversial. Stöcklin (1974, 1977) concluded that during the Palaeozoic Iran was an extension of the Arabian platform, and thus a part of Gondwana, possibly bordered by a Palaeo-Tethys in the north. These connections were reversed in the Mesozoic, when Central Iran became attached to the northern continent, but detached from the Gondwana mother continent by the Neo-Tethys. Brunn (1976), however, believes that Iran has always been an integral part of Eurasia, whereas Takin (1972) considers Central Iran to have been a microcontinent which drifted between Gondwana and Eurasia.

New geological evidence (Davoudzadeh and Schmidt 1982; Davoudzadeh and Weber-Diefenbach 1987) indicates that the North Iran suture which separates the North Alborz range from the South Alborz (Fig. 1) is the major suture of the Palaeozoic Tethys, separating the southern carbonate shelf from the Scytho-Turanian active margin. The area north of the North Iran suture was part of the Scytho-Turanian

plate during most of the Palaeozoic. The closing of the Palaeo-Tethys was followed by the Hercynian orogeny and was completed by the Early Cimmerian movements in the Middle Triassic. In SW Iran the Zagros belt is a major marginal basin of the Arabian plate. It is limited to the NE by the Main Zagros Thrust (Neo-Tethyan suture).

Previous palaeomagnetic investigations in Iran (Becker et al. 1973; Soffel et al. 1975, 1980, 1989, 1992; Davoudzadeh et al. 1981; Soffel and Förster 1983, 1984; Schmidt and Soffel 1983, 1984) were focussed on the construction of the Apparent Polar Wander Path (APWP) of the Central-East-Iran Microplate (CEIM). Wensink (Wensink 1979, 1982, 1983; Wensink and Varekamp 1980) also contributed to the palaeomagnetic database for Iran. The most recent review on this subject has been published by Soffel et al. (1992), and all previously published pole positions for the CEIM and the Turan Plate are listed in Table 4. All data in conjunction with the new results presented in this study have been used for the reconstruction of the drift history and rotations of the CEIM with respect to the neighbouring tectonic units (e.g. the Turan Plate in the north and Arabia in the south). The large differences between the APWP's of the CEIM, Gondwana and the Turan Plate were interpreted in terms of differences in palaeo-latitudes and relative rotations between the blocks. The possible phases of continental collisions (Kimmerian and Alpidic phases) could be demonstrated. The geological consequences of our palaeomagnetic studies have been discussed by Davoudzadeh et al. (1981) and Schmidt and Soffel (1984).

With the logistical support of the Geological Survey of Iran (GSI) it was possible to carry out additional palaeomagnetic sampling in 1986 and 1991–1992 on rocks dating from the Precambrian to the Tertiary. Part of these samples served for test measurements on oriented hand samples in new areas of interest to investigate the suitability of various rock types for later studies. Reliable new palaeomagnetic data could be obtained from Cretaceous and Triassic rocks. For details and other results we refer to Rolf (1991) and Schmidt (1992).

Major structural units of Iran

Fundamental differences in crustal character and in age of basement consolidation allow three major structural units to be distinguished in Iran, separated from each other by ophiolite bearing sutures. The three major units and their main constituents are as follows:

1. The southern unit with a crystalline basement which consolidated in Late Precambrian times, and a platform-type Palaeozoic development with continuous sedimentation from Triassic to Plio-Pleistocene times. This unit comprises the Zagros folded belt, which was folded in a late alpine tectonic phase (Pliocene).
2. The central unit is interpreted as an assemblage of marginal Gondwana fragments that were united with the mother continent and separated from the northern continent (Eurasia) in the Palaeozoic, but detached from Gondwana and attached to Eurasia in the Early Mesozoic, and were finally rejoined by Gondwanic Afro-Arabia in the Late Cretaceous–Early Tertiary. It comprises Central Iran and the Alborz. This unit went through all different alpine tectonic phases (Stöcklin 1977).
3. The northern unit is sharply separated from the central unit by the North Iran suture (Davoudzadeh and Schmidt 1982). It is characterized by continental crust including remnants of more or less cratonized former (Palaeozoic) oceanic crust which are thought to represent relics of the Palaeo-Tethys. The northern unit represents a marginal strip of the Hercynian realm of Central Asia, broadly overlapped by the Alpine realm. It was deformed and largely consolidated by strong early Kimmerian folding and a Late Alpine folding. The northern unit comprises the Talesh Range (NW of Rasht), northern Alborz, the South Caspian depression and the Kopet Dagh Range.

Other criteria, such as structural style, age and intensity of deformation, as well as age and nature of magmatism, are used to subdivide these major zones into smaller structural–stratigraphic blocks.

Geology and structural history of the Central-East-Iran microplate

Central Iran, in a broad sense, comprises the whole area between the North and South Iranian range. Within the Iran Plate the Central-East-Iran microplate is bordered by the Great Kavir fault in the north, by the Nain-Baft fault in the west and southwest, and by the Harirud fault in the east (Fig. 1). It is surrounded by the Upper Cretaceous to Lower Eocene ophiolite and ophiolitic mélange. The microplate itself consists of different structural components: the Lut Block, the Kerman-Tabas Block, the Yazd Block and the Anarak-Khur Block. All these blocks are separated from each other by major faults and have different lithostratigraphic records. The major stratigraphic, tectonic and magmatic characteristics of the different blocks are as follows:

The *Lut Block* was consolidated in Upper Proterozoic. Its younger sedimentary cover is generally flat lying, and large areas are covered by Cainozoic volcanics and Quaternary sand dunes and playas.

The *Kerman–Tabas Block* is characterized by strong lateral variations in thickness of the Palaeozoic formations (up to 7000 m in the Tabas–Shirgesht area and only 1000 m in the Kalmard area NW of Tabas) and by well-developed Mesozoic sedimentary strata. Cenozoic volcanics, widely distributed in the Lut Block, are almost totally absent in the Kerman–Tabas Block.

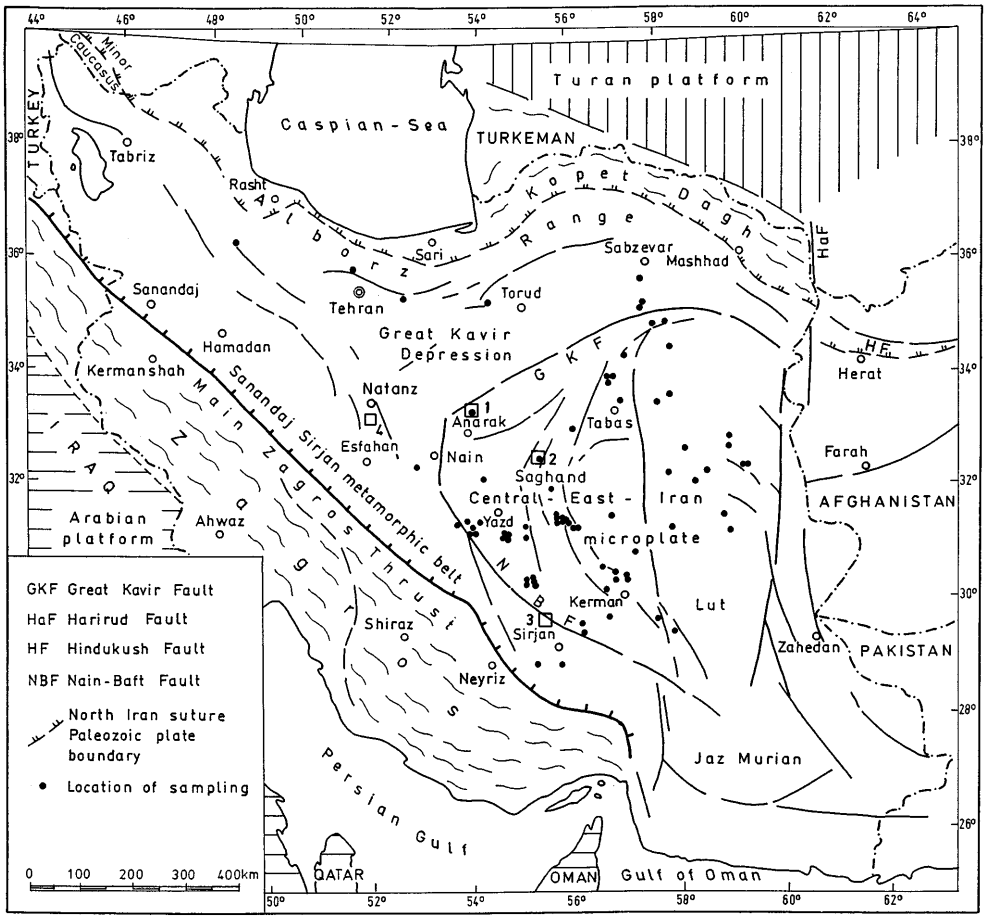


Fig. 1 Structural map of Iran showing the Central-East-Iran microplate (CEIM), its major fault systems and the locations of the sampling areas (open squares). *1* Area of Nakhlak; *2* area of Saghand; *3* area of Sirjan; *4* area of Natanz. *Open circles* major cities; *closed circles* palaeomagnetic sampling sites of previous investigations

The *Yazd Block*, in contrast to the other blocks, has Lower Palaeozoic formations developed mainly in evaporite facies. Middle and Upper Jurassic formations which are well developed in the Tabas–Kerman Block, are missing or only poorly developed in the Yazd Block. Mesozoic magmatism is common (Shir-Kuh Granite 50 km SW of Yazd, Zarin Granite 80 km NNE of Yazd).

The *Anarak–Khur Block* is characterized by a thick (5000–8000 m) Palaeozoic metamorphic series consisting mainly of detrital sediments and volcanic rocks. The marine Triassic sequence of Nakhlak (up to 2500 m thick) is significantly different from the carbonate platform facies of Central Iran. Cretaceous formations are well developed (Lower Cretaceous is up to 3000 m thick). The Saghand–Bafq Block belongs to this Block; it was dislocated later tectonically towards the south (Davoudzadeh and Weber-Diefenbach 1987).

Sampling areas and measurement procedures

The sampling areas are indicated in Fig. 1. Area no. 1 is situated in the Anarak–Khur Block near Nakhlak and is part of the CEIM. Area no. 2 is located near Saghand and lies between the Yazd Block and the Kerman–Tabas Block, both of which belong to the CEIM. Area no. 3 to the north of Sirjan is situated at the southern rim of the CEIM. Clearly outside the CEIM is area no. 4 in the region of Natanz. Table 1 shows the sites, geographic coordinates, lithologies, rock ages, number of sites and number of samples. The tectonic setting of the sites was also determined for tectonic correction to the palaeohorizontal position and for a fold test.

Drill cores were taken from the in situ rock formation or from oriented hand samples, from which standard cylindrical specimens (2.5 cm in diameter, 2.2 cm in length) were obtained. The measurement and the analysis of the natural remanent magnetization (NRM) was done mostly with a superconducting (cryogenic) magnetometer of type LETI (Grenoble, France). The NRM of the rocks was tested with respect to its stability and origin using alternating field (AF) and/or thermal (TH) demagnetization, depending on the predominant ferromagnetic ores (magnetite, titanomagnetite, hematite, goethite) of the rocks. The decay of the NRM during the demagnetization treatment in field-free space was analysed using multicomponent analysis for the determination of the characteristic remanent magnetization (ChRM) as decribed in standard textbooks on palaeomagnetism (e.g. Van der Voo 1993). Both orthogonal vector plots (Zijderveld diagrams) and equal area stereonets were used for the presentation of the analysis of the NRM. Additional rock magnetic measurements

Table 1 List of the new palaeomagnetic sampling areas in Iran. Latitude in °N Longitude in °E

Sampling area no.	Area name	Latitude (N)	Longitude (E)	Lithology	Age	Sites (*n*)	Samples (*n*)
1	Nakhlak	33.4	53.9	Limestone, tuff	Lower–Middle Triassic	7	92
2	NW Saghand	32.5	55.2	Limestone	Lower Cretaceous	4	112
3	Sirjan	29.4	55.3	Basalts	Lower–Middle Triassic	2	89
4	Natanz	33.2	51.7	Dolomite Limestone Sandstone Siltstone	Middle Triassic Triassic Triassic	5	44

Table 2 Palaeomagnetic results. For sampling area number, name and rock ages, see Table 1. Pole number and palaeomagnetic technique. AF: Alternating field demagnetization. TH: Thermal demagnetization. N: number of samples. Declination D and inclination I of characteristic remanent magnetization ChRM. α_{95} is the radius of the cone of confidence; Virtual geomagnetic pole position; Palaeolatitude ϕ_{pal}

Area no.	Area name	Pole no.	Age	*N*	ChRM		α_{95} (°)	Pole position		ϕ_{pal} (°N)
					D (°E)	I (°)		(°E)	(°N)	
1	Nakhlak	1 (TH)	Triassic	12	94.0	25.0	24.7	310.8	− 3.9	13.1
2	Saghand	(AF)	Cretaceous	16	345.2	28.5	4.3	276.5	68.1	15.2
2	Saghand	(TH)	Cretaceous	17	338.5	25.2	5.2	285.8	62.5	13.2
2	Saghand	2[a]	Cretaceous	33[a]	340.7	26.3	3.8	283.1	64.4	13.9
3	Sirjan	(AFe)	Triassic	25	113.7	34.8	5.0	296.7	9.8	19.2
3	Sirjan	(TH)	Triassic	19	115.5	35.4	3.9	295.3	10.9	19.6
3	Sirjan	3[a]	Triassic	44[a]	114.5	35.1	3.2	295.8	10.3	19.4
4	Natanz	4[b] (TH)	Triassic	5[b]	33.3	25.1	9.3	167.2	53.7	13.2

[a] Combination of all ChRM directions obtained after TH and AF demagnetization
[b] Mean of five sites with 161 specimens

(IRM acquisition curves and their demagnetization, Curie temperature measurement and ore microscopy) were also made to determine the carrier of the most stable component of the NRM. The ChRM of the rocks determined in this paper are listed in Table 2. They include tectonic correction, if not otherwise stated.

Palaeomagnetic results

Nakhlak (area no. 1)

The Nakhlak area (part of the Anarak–Khur Block) is situated approximately 30 km northeast of the small town of Anarak in the southern part of the Great Kavir (Fig. 1). The Triassic Nakhlak group, described by Davoudzadeh and Seyed-Emami (1972), comprises approximately 2540 m of limestone, sandstone, shale and conglomerate . Much of the rock is tuffaceous. The rocks of the Nakhlak group range in age from Upper Scythian to Ladinian (Early and Middle Triassic). The base of the Nakhlak group is not exposed; dune deposits cover the lowest exposures. The sequence is overlain unconformably by limestones and sandstones of Cretaceous age. The marine Triassic sequence of Nakhlak is significantly different from the carbonate platform facies of Central Iran. In Iran this facies type is exposed only in a few localities north of the North Iran suture (e.g. Aghdarband in eastern Kopet Dagh). The Nakhlak group was displaced from its original position (west of Herat on the Turan plate) by the rotation of the Central-East-Iran microplate (Davoudzadeh et al. 1981).

In this area we collected hand samples from Triassic limestones at three sites and tuffites at four sites from which 92 specimens were obtained. The mean initial in situ NRM direction of the limestones is: D = 221.6°, I = 86.7°, N = 42, k = 9.8, α_{95} = 7.4° and for the tuffites: D = 340.6°, I = 66.4°, N = 50, k = 8.3, α_{95} = 7.4°. They differ significantly from the present local geomagnetic field direction in Iran (D = 5°, I = 55°). The AF demagnetization was not successful for the complete destruction of the NRM. Only few specimens showed a consistent grouping of directions during thermal demagnetization (see Fig. 2a, b) in the blocking temperature range above 300 °C after bedding correction. The mean ChRM direction of 12 specimens from four sites is:

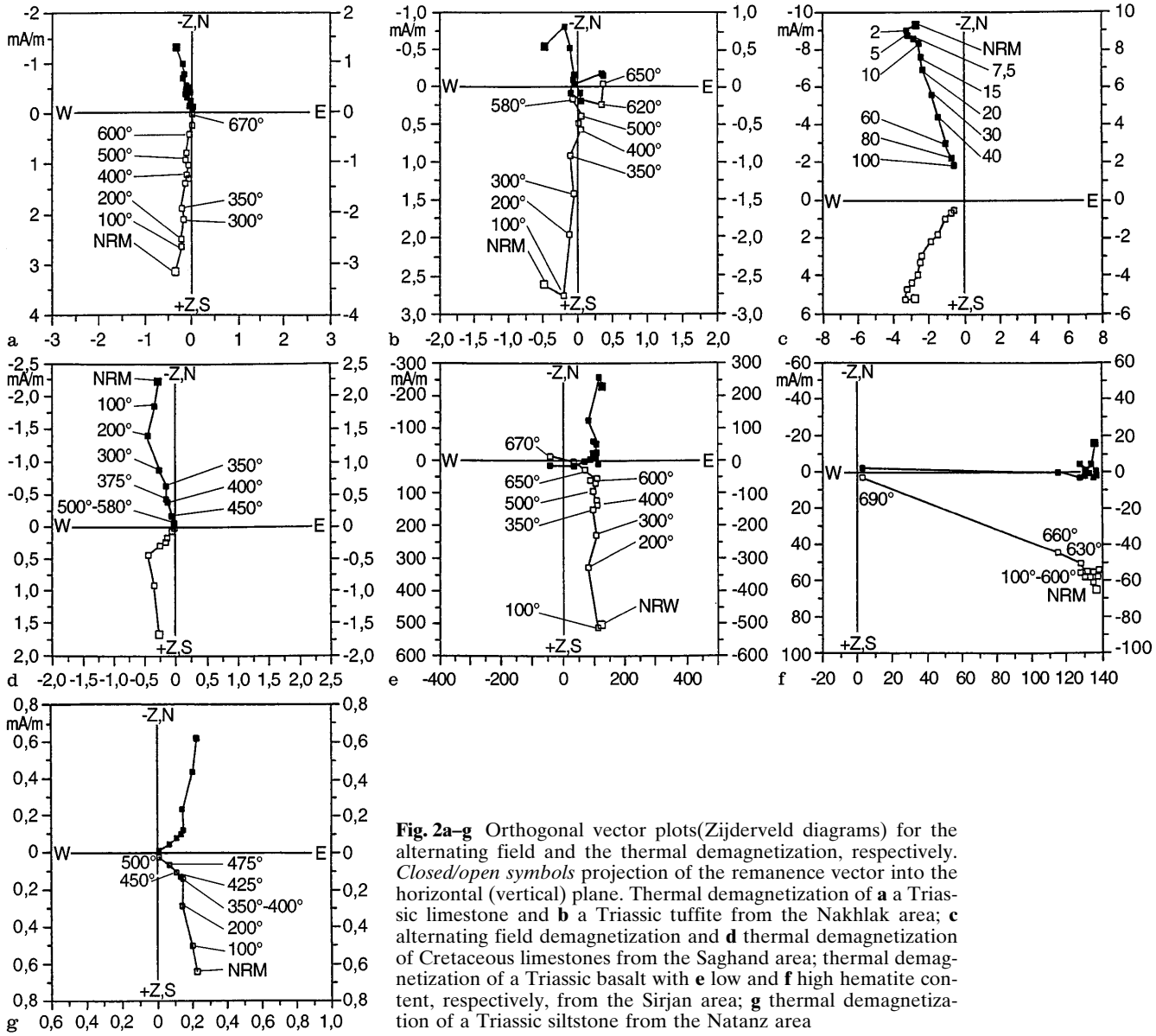


Fig. 2a–g Orthogonal vector plots(Zijderveld diagrams) for the alternating field and the thermal demagnetization, respectively. *Closed/open symbols* projection of the remanence vector into the horizontal (vertical) plane. Thermal demagnetization of **a** a Triassic limestone and **b** a Triassic tuffite from the Nakhlak area; **c** alternating field demagnetization and **d** thermal demagnetization of Cretaceous limestones from the Saghand area; thermal demagnetization of a Triassic basalt with **e** low and **f** high hematite content, respectively, from the Sirjan area; **g** thermal demagnetization of a Triassic siltstone from the Natanz area

D = 94.0°, I = 25.0°, N = 12, k = 4.1, α_{95} = 24.7°. The small and unevenly distributed number of useful ChRM directions per site did not allow to calculate individual site mean directions. The mean ChRM direction and its virtual geomagnetic pole are listed in Table 2. The corresponding VGP (pole no. 1 in Fig. 4) is discussed later.

Saghand (area no. 2)

The Saghand area is situated on the eastern margin of the Yazd Block in the centre of Central Iran (Fig. 1) and forms part of the Central-East-Iran microplate. In this area several sequences of Palaeozoic metamorphic rocks are exposed. The rocks vary from high-grade metamorphic to slightly and even nonmetamorphosed varieties. All are associated with volcanics and/or volcano-detritics. The Saghand area is dominated strongly by N-S trending basin and range extensional tectonics, forming a mosaic structure of mainly N-S elongated zones. The faults between them have been reactivated at various times.

The Cretaceous limestones are slightly metamorphic. A number of 112 samples (cores) were collected within the same more or less subhorizontal formation mainly at four sites along a profile of 30 km. The in situ NRM groups around a mean direction of D = 346.5°, I = 35.4°, N = 112, k = 9.8, α_{95} = 4.5°. Only few samples gave reliable results after the minor bedding corrections. With AF demagnetization a consistent ChRM(AF) component could be isolated between 10 mT and 100 mT on 16 samples with a mean direction of D = 345.2°, I = 28.5°, N = 16, k = 74.0, α_{95} = 4.3°. By

thermal demagnetization a ChRM(TH) component with D = 338.5°, I = 25.2°, N = 17, k = 47.0, α_{95} = 5.2° could be isolated in the blocking temperature range between 300 and 590 °C. The corresponding Zijderveld diagrams are shown in Fig. 2c and d. Both ChRM directions are regarded as primary. The combined mean remanence direction obtained from alternating field and thermal demagnetization (D = 340.7°, I = 26.3°, N = 33, k = 44.3, α_{95} = 3.8°) and its virtual geomagnetic pole (pole no. 2) are listed in Table 2. The VGP is also plotted in Fig. 4.

Sirjan (area no. 3)

The Sanandaj–Sirjan metamorphic belt, which runs parallel to the Zagros from SE to NW Iran, joins the Taurus orogenic belt in Turkey (Fig. 1). This belt is separated from the Zagros by a narrow zone of deep thrust and reverse faults (the Zagros Thrust Zone).

In the Sirjan area, at the western side of the Nain–Baft fault, the Palaeozoic metamorphic sequence is more than 5000 m thick. It consists of carbonates, pelitic and psammitic rocks as well as basic volcanics and ultrabasic rocks. K/Ar radiometric dating of the Ghouri area (west of Sirjan) revealed ages of 404 ± 8 Ma to 362 ± 7 Ma (Adib 1978). Devonian pollen and spores, some Crinoids and other fossil fragments are reported as well.

The Triassic basalts in the Sirjan area are only slightly metamorphic. The initial NRM directions of the 89 samples (cores) from two sites are very scattered. After AF demagnetization two groups of stable directions could be found. One of them, called ChRM(AF1) has a mean in situ direction (before tectonic correction) of D = 3.3°, I = 47.9°, N = 27, k = 51.3, α_{95} = 3.9° which points more or less in the direction of the present local geomagnetic field in Iran (D = 5°, I = 55°). Although 5 of the 22 directions are reversed (passing a positive reversal test), we think that this component is secondary because of its coincidence with typical Tertiary remanence directions in Iran. Its VGP (153.2 °E, 87.1 °N) is close to the present geographic pole and agrees with Late Tertiary pole postions of the CEIM. The other group of directions after AF demagnetization in the range between 10 and 100 mT has a mean direction after bedding correction of D = 113.7°, I = 34.8°, N = 25, k = 34.5, α_{95} = 5.0° and is called ChRM(AF2). The ChRM(AF2) direction was also confirmed by thermal demagnetization of a second set of samples in the blocking temperature range from 400 to 590 °C yielding a ChRM(TH) with D = 115.5°, I = 35.4°, N = 19, k = 76.0, α_{95} = 3.9°. Figure 2e and f show the corresponding Zijderveld diagrams. Two of the 19 directions are reversed passing a positive reversal test. A combination of the ChRM(AF2) and the ChRM(TH) directions yields a mean ChRM direction of D = 114.5°, I = 35.1°, N = 44, k = 45.9, α_{95} = 3.2°. The corresponding VGP (pole no. 3) is shown in Fig. 4. It is situated close to other Late Palaeozoic/Early Mesozoic VGP positions of the CEIM. We regard the combined ChRM(AF2)/ChRM(TH) directions as a primary Triassic remanence component. All data are listed in Table 2.

Natanz (area no. 4)

The Natanz–Soh area is situated between Esfahan and Kashan (Fig. 1). It forms part of the western Central Iran, but is situated outside of the CEIM (almost 150 km to the west). The geology of this area has been described by Zahedi (1973).

The area contains no evidence of Carboniferous and Early Permian beds. This important gap displays no angular unconformity. Permian deposits are conformably overlying the last beds of the Devonian Bahram Formation. The Triassic beds begin with 50 m of red conglomerates, sandstones and siltstones (Abyaneh Formation) which is attributed to the Lower Triassic (Upper Scythian). On top of this detritic sequence follow 1000 m of dolomites and limestones corresponding to the Middle Triassic Shotori Formation.

There is a very rapid change in the nature of the sedimentation from Middle Triassic dolomites and limestone to the very thick, sandy argillaceous deposits with some limestones of the Upper Triassic (Nayband Formation, 1400 m) and Liassic (Shemshak Formation, 1000 m). This is regarded as evidence for a change in the palaeogeographic conditions during the Triassic. No evidence of the Dogger, Malm and Neocomian stages could be found in the Natanz–Soh area. The Aptian and Albian limestones and marls (1000 m) unconformably overly folded beds of the Liassic and Upper Triassic.

A number of 44 drill cores were collected at five sites yielding 161 specimens of red sandstones and siltstones of the Abyaneh Formation, dolomites of the Shotori Formation (Lower-Middle Triassic) and limestones of the Nayband Formation (Upper Triassic). Only thermal demagnetization was successful for the analysis of the NRM. A Zijderveld diagram for thermal demagnetization is shown in Fig. 2g. From all five sites well-de-

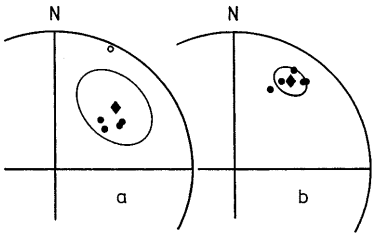


Fig. 3a, b Equal area projection of the mean characteristic remanence directions of the five sites of Triassic rocks of the Natanz area. *Closed/open symbols* positive (negative) inclination. **a** Mean directions of all five sites before tectonic correction; **b** mean directions of all five sites after tectonic correction showing the reduction of the scatter and the positive fold test

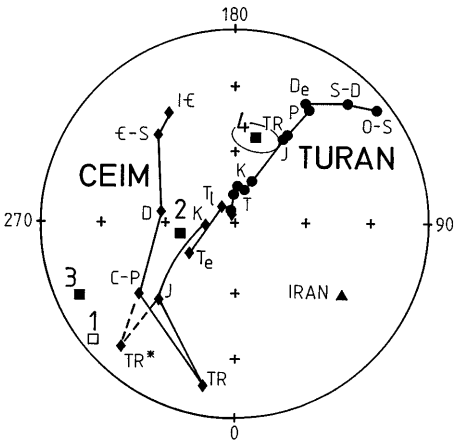


Fig. 4 Apparent Polar Wander Path (north poles) of the Central-East-Iran microplate (CEIM; *diamonds*) after Soffel et al. (1992) and of the Turan Plate (*circles*) after Khramov and Rodionov (1980) together with the new pole positions of this investigation (*squares*). For pole numbers see Table 2. *Closed/open symbols* poles on the northern (southern) hemisphere. *Triangle* mean geographic coordinates of Iran. Conventional abbreviations have been used for the geological formations using the subscripts *l* for Late and *e* for Early, respectively. (See text for further details)

fined ChRM directions could be determined from a sufficient and comparable number of specimens. The mean ChRM directions of each site are also plotted in Fig. 3 before and after tectonic correction. At most sites two polarities of remanence could be observed, and in all cases the reversal test of McFadden and McElhinny (1990) is seen to be positive. Before tectonic correction the five sites have the following mean ChRM direction of D = 44.6°, I = 37.5°, N = 5, k = 10.5, α_{95} = 24.8°. After tectonic correction, there is only a small directional shift of the mean (D = 33.3°, I = 25.1°, N = 5, k = 69.0, α_{95} = 9.3°), but the scatter is reduced considerably. The precision parameter k is improved after the tectonic correction by a factor of 6.6, which means that the fold test is positive with a probability of 99% using the fold test of McElhinny (1964). Therefore, we regard this ChRM direction as a primary Triassic direction. The mean remanence direction of the five sites and the virtual geomagnetic pole position are listed in Table 2. The corresponding VGP (pole no. 4) is plotted in Fig. 4. It is situated close to the mean Triassic pole of the Turan Plate.

Interpretation of pole positions

Figure 4 shows the Apparent Polar Wander Paths (APWP; north poles) of the CEIM (diamonds) and the Turan Plate (circles) using the data listed in Table 4 together with the new pole positions of this paper (squares). Their numbers refer to the pole numbers given in Table 2. The mean location of the CEIM (58°E; 32°N) is indicated by the triangle.

The Triassic pole (no. 1) from the Nakhlak area is, within the error limits, in reasonable agreement with the previously published Triassic pole of Soffel et al. (1984); see also Table 3. It confirms in principle the large counterclockwise rotation of the CEIM since the Triassic (Davoudzadeh et al. 1981; Schmidt and Soffel 1983). The Cretaceous pole (no. 2) from the Saghand area is regarded as reliable from the demagnetization behaviour of the rocks and several rock magnetic characteristics indicating a primary carrier of remanence. It is close to other Cretaceous poles of the CEIM and confirms this section of its APWP. In the Sirjan area on the southern rim of the CEIM the Triassic pole (no. 3) is situated in the vicinity of other Late Palaeozoic–Early Mesozoic poles of the CEIM and is far apart from the Triassic pole position for the Turan Plate. From this we conclude that the Sirjan area may be regarded as part of the CEIM. The area to the north of Sirjan must also have carried out a considerable counterclockwise rotation (ca. 120°) with respect to the Turan Plate since the Triassic.

Table 3 Compilation of Triassic poles for the CEIM, the Iran Plate (area of Natanz outside the CEIM), Gondwana and the Turan Plate. dp: error of the colatitude. dm: error perpendicular to dp

Plate	CEIM pole position		Iran Plate pole position		Gondwana pole position		Turan plate pole position	
	(°E)	(°N)	(°E)	(°N)	(°E)	(°N)	(°E)	(°N)
Pole no. 1, this paper	310.8	− 3.9						
Pole no. 4, this paper	295.8	10.3						
Soffel et al. (1992)	338.9	20.7						
Wensink (1982)	326.2	21.6						
This paper			167.6	53.6				
Van der Voo (1993)					260	56		
Khramov and Rodionov (1980)							149	49
Mean values:	317.5 N = 4 areas α_{95} = 25.3°	12.7	167.6 N = 5 sites dp = 5.4°; dm = 10.0°	53.6	260	56	149	49

The new more reliable Triassic pole for the CEIM based on poles from four different areas in the CEIM has the following data: 317.5°E, 12.7°N, N = 4 areas, K = 14.1, α_{95} = 25.3°. It is plotted in Fig. 4 as a diamond (TR*) and has been connected by dashed lines with the APWP of the CEIM as an alternative path for the Triassic. The extended sampling of Triassic rocks in the Natanz area (pole no. 4) gave reliable data according to the standard criteria in palaeomagnetism (sufficient number of samples, positive reversal test and fold test). The close agreement of this Triassic VGP with the APWP of the Turan Plate (see Fig. 4) indicates that the Natanz region, and possibly also the entire area to the west of the CEIM (between the Alborz in the north and the Zagros in the south), must be regarded as being situated close to the southern extension of the Turan Plate during the Triassic. The presently available Triassic poles for the CEIM and for the area of Natanz outside the CEIM have been compiled in Table 3.

Conclusions and geodynamical interpretation

The new palaeomagnetic investigations in Iran were extended also to some areas outside the Central-East-Iran microplate and can be summarized as follows: The APWP of the CEIM from the Triassic through the Tertiary is generally confirmed by new data from areas within or close to the CEIM. Triassic poles from volcanic rocks of the area north of Sirjan indicate that this area must also have been rotated together with the CEIM. The new mean Triassic pole for the CEIM, derived from the results of Soffel et al. (1992), Wensink (1983) and this paper (see Table 3) is at: 317.5°E, 12.7°N, N = 4 areas, K = 14.1, α_{95} = 25.3°. It has been named TR* in Fig. 4 and has been connected with the APWP of the CEIM with dashed lines. Because the four Triassic poles from the CEIM originate from areas close to major fault systems, we think that their scatter is due to small local tectonic movements.

The Triassic pole (no. 4) from the Natanz area outside the CEIM (167.2°E, 53.7°N, N = 5 sites, dm = 10.0°, dp = 5.4) is in good agreement with the mean Triassic pole of the Turan plate (149°E; 49°N) showing that the area to the west of the CEIM was close to the Turan Plate during the Triassic.

Previously existing palaeomagnetic data for the CEIM (Soffel et al. 1992; listed in Table 4) and the data of this paper were used to reconstruct the palaeocontinental configuration of the region for the Triassic using published data from Gondwana (Van der Voo 1993), the Turan Plate (Khramov and Rodionov 1980) and the new results for the Triassic of this paper (see Table 3). The reconstruction is shown in Fig. 5.

The present configuration of the plates and microplates in the vicinity of Iran are shown in Fig. 5a. The CEIM has been marked with a north-pointing arrow for a better illustration of its rotation since the Triassic. Figure 5b represents the situation in the Triassic (220 Ma). During the Triassic the Iran Plate (without the CEIM, which is regarded as a separate unit) is situated close to the rims of the southern extension of the Turan Plate, whereas the CEIM is in a more southeasterly position. Its progressive westward motion was combined by a large (120–130°) counterclockwise rotation.

Table 4 Summary of pole position from the Central-East-Iran microplate and the Turan Plate according to Soffel et al. (1992) and Khramov and Rodionov (1980), respectively

	Central-East-Iran microplate		Turan plate	
Age formation	Virtual pole position (°E)	(°N)	Virtual pole position (°E)	(°N)
Late Tertiary	215.0	81.6	198	85
Middle Tertiary			180	78
Early Tertiary	303.6	66.9	175	75
Cretaceous	273.3	78.9	157	71
Jurassic	314.7	43.2	163	76
Triassic	348.9	15.3	149	49
Permo-Triassic			149	48
Early Permian			146	31
Carboniferous to Permian	306.5	38.1		
Devonian	261.8	58.4		
Middle to Late Devonian			149	29
Silurian to Early Devonian			136	16
Ordovician to Silurian			128	6
Cambrian to Silurian	221.8	39.7		
Infracambrian	210.8	34.4		

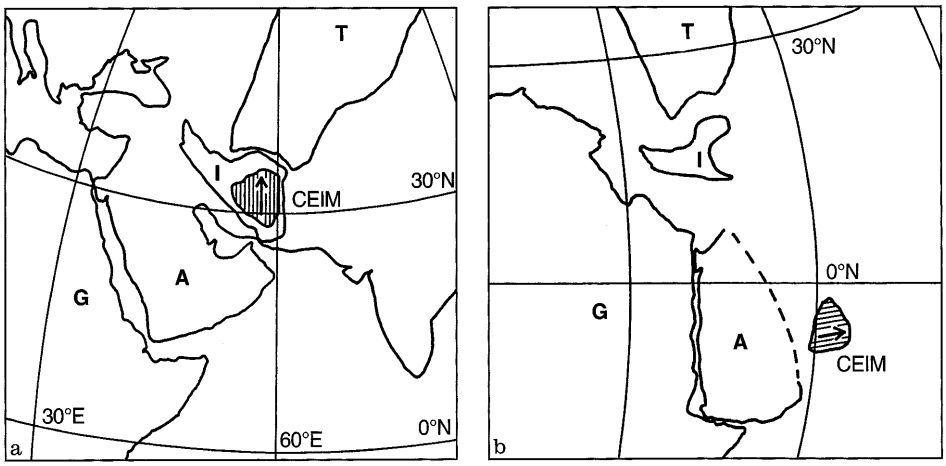


Fig. 5a, b Palaeocontinental reconstruction for Gondwana (*G*) including Arabia (*A*), the Turan Plate (*T*) and the Iran Plate (*I*), of which the Central-East-Iran microplate (*CEIM*) forms a separate block. *Arrow* present north-pointing direction within the CEIM. **a** Present situation; **b** reconstruction for the Middle–Upper Triassic (220 Ma). During this time the CEIM formed a separate unit. It was attached to the Iran Plate in Late Mesozoic–Early Tertiary times

The CEIM performed an anticlockwise rotation of nearly 65° from the Late Triassic to Middle Jurassic. During the Upper Jurassic and the Cretaceous, the CEIM carried out a further anticlockwise rotation of 35°. The rift zones were closed in Early Tertiary by convergence of the CEIM and Iran plate. This collision was accompanied by additional anticlockwise rotation of the CEIM of 35°. Thus, the total amount of anticlockwise rotation of the CEIM since the Triassic with respect to Eurasia is approximately 135°. Ophiolite and ophiolitic mélange surrounding the Central-East-Iran microplate are the remnants of this rift zone.

Acknowledgements The authors are indebted to the Geological Survey of Iran for support during the sampling trips. The financial support of the "Deutsche Forschungsgemeinschaft" (DFG) is also gratefully acknowledged. Dr. V. Bachtadse helped to make the palaeocontinental reconstruction, which was performed using the GMAP software by Dr. T. Torsvik. We are also grateful for the helpful comments by Dr. H. Wensink, from Utrecht, and Dr. J. Tait, in the reviewing process.

References

Adib D (1978) Geology of the metamorphic complex at the south-western margin of the Central-East Iran microplate (Neyriz area). N Jb Geol Paläont Abh 156:393–409

Becker H, Förster H, Soffel HC (1973) Central Iran, a former part of Gondwanaland? Palaeomagnetic evidence from Infracambrian rocks and iron ores of the Bafq area, Central Iran. Z Geophys 39:953–963

Brunn JH (1976) Über die Entstehung gefalteter Ketten: Kollisions-Tektonik und induzierte Bögen. Z Dtsch Geol Ges 127:323–335

Davoudzadeh M, Seyed-Emami K (1972) Stratigraphy of the Triassic Nakhlak Group Anarak region, Central Iran. Geol Surv Iran Rep 28:5–28

Davoudzadeh M, Schmidt K (1982) Zur Trias des Iran. Geol Rundsch 71:1021–1039

Davoudzadeh M, Weber-Diefenbach K (1987) Contribution to the paleogeography, stratigraphy and tectonics of the Upper Paleozoic of Iran. N Jahrb Geol Paläont Abh 175:121–145

Davoudzadeh M, Soffel HC, Schmidt K (1981) On the rotation of the Central-East-Iran microplate. N Jahrb Geol Paläontol Mh 3:180–192

Khramov AN, Rodionov VP (1980) Palaeomagnetism and reconstruction of the palaeogeographic positions of the Siberian and Russian Plates during the Late Proterozoic and Palaeozoic. J Geomagn Geol 32 (Suppl III):23–37

McElhinny MW (1964) Statistical significance of the fold test in Paleomagnetism. Geophys J R Astr Soc 8:338–340

McFadden P, McElhinny MW (1990) Classification of the reversal test in palaeomagnetism. Geophys J Int 103:725–729

Rolf C (1991) Paläo- und gesteinsmagnetische Untersuchungen an Proben aus dem Zentraliran und Reinterpretation paläomagnetischer Ergebnisse aus der Zentral-Ost-Iran Mikroplatte (CEIM) unter Berücksichtigung geologischer Modelle. Dissertation, Fakultät für Geowissenschaften, Universität München, pp 1–125

Schmidt S (1992) Paläomagnetische und gesteinsmagnetische Untersuchungen an Gesteinen aus dem NW Iran. MSc. Thesis, Institut für Allgemeine und Angewandte Geophysik, Universität München, pp 1–110

Schmidt K, Soffel HC (1983) Mesozoic-Cenozoic geological events in Central-East-Iran and their relation to palaeomagnetic results. Geol Surv Iran Rep 51:27–35

Schmidt K, Soffel HC (1984) Mesozoic-Cenozoic geological events in Central-East-Iran and their relation to palaeomagnetic results. N Jahrb Geol Paläontol Abh 168:173–181

Soffel HC, Förster HG (1980) Apparent polar wander path of Central Iran and its geotectonic interpretation. J Geomagn Geoelectr (Suppl III):117–135

Soffel HC, Förster HG (1983) Polar wander path of the Central-East-Iran microplate including new results. Geol Surv Iran Rep 51:13–26

Soffel HC, Förster HG (1984) Polar wander path of the Central-East-Iran microplate including new results. N Jahrb Geol Paläontol Abh 168:165–172

Soffel HC, Förster H, Becker H (1975) Preliminary polar wander path of Central Iran. J Geophys 41:541–543

Soffel HC, Davoudzadeh M, Rolf C (1989) Palaeomagnetic investigations on Phanerozoic formations from Iran: reinterpretation of measurements between 1972 and 1982. Münchner Geophys Mitt Münchner Univers Schriften 4:23–56

Soffel HC, Davoudzadeh M, Rolf C (1992) A New Polar Wander Path of the Central East Iran Microplate (CEIM) and its tectonic interpretation. Geosci Sci Q J Geol Surv Iran 2 (5):2–24

Stöcklin J (1974) Possible ancient continental margins in Iran. In: Burk CA, Drake CL (eds) The geology of continental margins, pp 873–887

Stöcklin J (1977) Structural correlation of the Alpine ranges between Iran and Central Asia. Mem Soc Geol France Ser H 8:333–353

Takin M (1972) Iranian geology and continental drift in the Middle East. Nature 235 (5334):147–150

Van der Voo R (1993) Palaeomagnetism of the Atlantic, Tethys and Iapetus Oceans. Cambridge University Press, Cambridge
Wensink H (1979) The implication of some palaeomagnetic data from Iran for its structural history. Geol Mynbouw 58:175–185
Wensink H (1982) Tectonic inferences of palaeomagnetic data from some Mesozoic formations in Central Iran. J Geophys 51:12–23
Wensink H (1983) Palaeomagnetism of Red Beds of Early Devonian age from Central Iran. Earth Planet Sci Lett 63:325–334
Wensink H, Varekamp JC (1980) Palaeomagnetism of basalts from Alborz: Iran part of Asia in the Cretaceous. Tectonophysics 68:113–129
Zahedi M (1973) Etude géologique de la région de Soh (W de l'Iran Central). Geol Surv Iran Rep 27:1–197